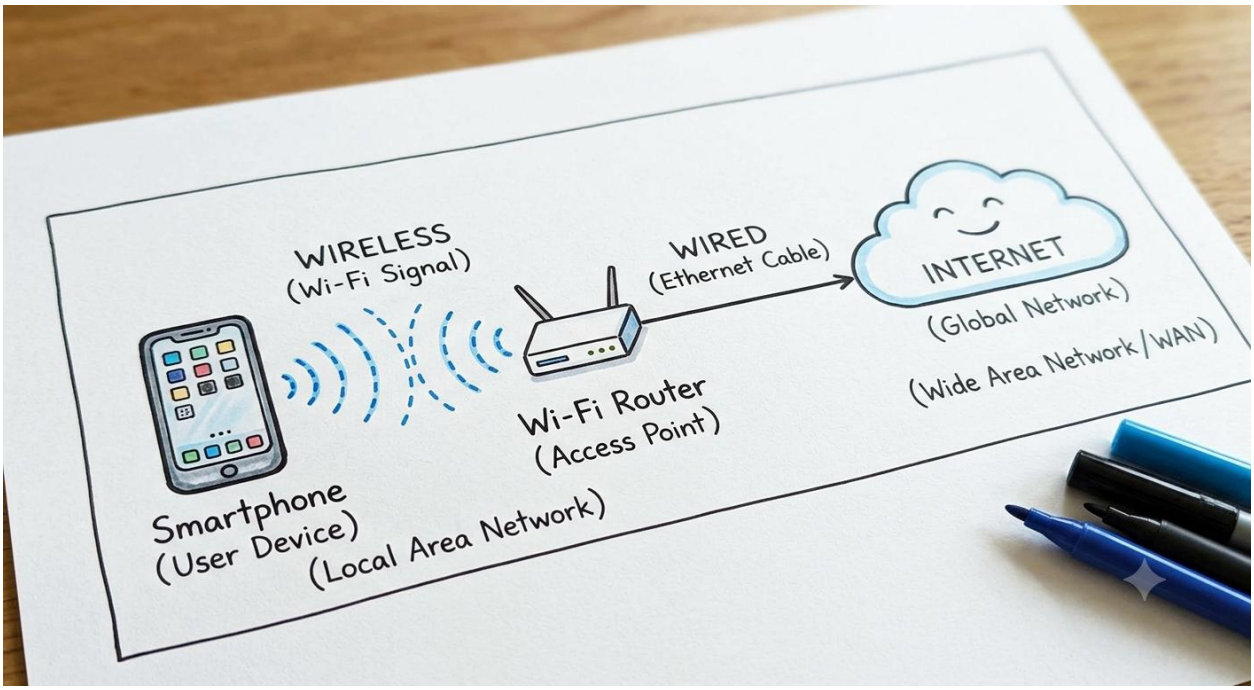
1. **List 3 popular Wi-Fi standards (e.g., 802.11n, 802.11ac, 802.11ax) and for each, write one key difference in speed or frequency band.**

Wi-Fi Standard	Frequency Band	Key Difference
802.11n (Wi-Fi 4)	2.4 GHz / 5 GHz	Speed up to 600 Mbps
802.11ac (Wi-Fi 5)	5 GHz	Faster, up to about 6.9 Gbps
802.11ax (Wi-Fi 6)	2.4 GHz / 5 GHz	Higher efficiency and speed, up to about 9.6 Gbps

2. **Draw a simple diagram (on paper or using any drawing tool) showing how your smartphone connects to the internet via a Wi-Fi router at home, labeling the wireless and wired parts of the network. generate an image**



3. **Scan for available Wi-Fi networks using your laptop or phone, note down the SSID, signal strength, and security type (WPA2, WPA3, etc.) for at least 3 networks, and explain which one you would choose and why. Hint: Consider both signal strength and security when choosing.**

❖ **Wi-Fi Network Scan**

SSID	**Signal Strength**	**Security**
Home_WiFi	Excellent (4/5)	WPA2
JioFiber_5G	Good (3/5)	WPA2/WPA3
TP-Link_5G	Fair (2/5)	WPA2

➢ **My Choice**

I would choose **JioFiber_5G** because it has a good signal and uses **WPA2/WPA3** security, which makes it a **safer** choice than an open or weakly secured network.

4. Explain in your own words what 'SSID hiding' is in Wi-Fi networks and discuss one advantage and one disadvantage of hiding your Wi-Fi SSID.

❖ **SSID hiding means making your Wi-Fi network name invisible when people scan for nearby Wi-Fi networks. The network is still there, but its name does not normally appear in the Wi-Fi list.**

- **Advantage:** It can make the network less noticeable to people nearby.

- **Disadvantage:** It does not provide **real security**. Someone can **still detect** the network using **Wi-Fi scanning tools**, and devices that connect to it may reveal its name.